

Q. LCS (Longest common sub-sequence)

Given two strings. Find the length of LCS b/w the two strings.

$S_1 : A G G T A B \longrightarrow 4 \quad \{G T A B\}$
 $S_2 : G X T N A Y B$

$S_1 : D E M O C R A T \longrightarrow 3 \quad \{E C A\}$
 $S_2 : R E P U B L I C A N$

$S_1 : C A R T H O R S E \longrightarrow 3 \quad \{R H S\}$
 $S_2 : O R C H E S T R A$
 $\{R H R\}$
 $\{R H E\}$

$S_1 : A B C \longrightarrow 1 \quad \{O R E\}$
 $S_2 : C B A$
 $\{O R E\}$
 $\vdots$

$S_1 : A B B C D G F \longrightarrow \text{LCS} \left(\begin{matrix} S_1 : A B B C D G \\ S_2 : B A C D E G \end{matrix} \right) + F$

$\text{LCS} \left(\begin{matrix} S_1 : A B B C D + G F \\ S_2 : B A C D E \end{matrix} \right)$

$\max \left\{ \text{LCS} \left(\begin{matrix} S_1 : A B B C D \\ S_2 : B A C D \end{matrix} \right) \right.$

$\left. \text{LCS} \left(\begin{matrix} S_1 : A B B C \\ S_2 : B A C D E \end{matrix} \right) \right\} + G F$

$S_1 : A G G T A B$
 $S_2 : G X T N A Y B$

$LCS(S_1, i, S_2, j)$

if $(S_1[i] == S_2[j])$ {
 ret $1 + LCS(S_1, 4, S_2, 5)$

$S_1 : A G G T A$
 $S_2 : G X T N A Y$

$LCS(S_1, 3, S_2, 5)$

$\max(LCS(S_1, 2, S_2, 5), LCS(S_1, 3, S_2, 4))$

$LCS(S_1, 4, S_2, 4)$

$1 + LCS(S_1, 3, S_2, 3)$

$LCS(S_1, i, S_2, j) \rightarrow$ Return length of LCS b/w
 $S_1 [0, i]$
 $S_2 [0, j]$

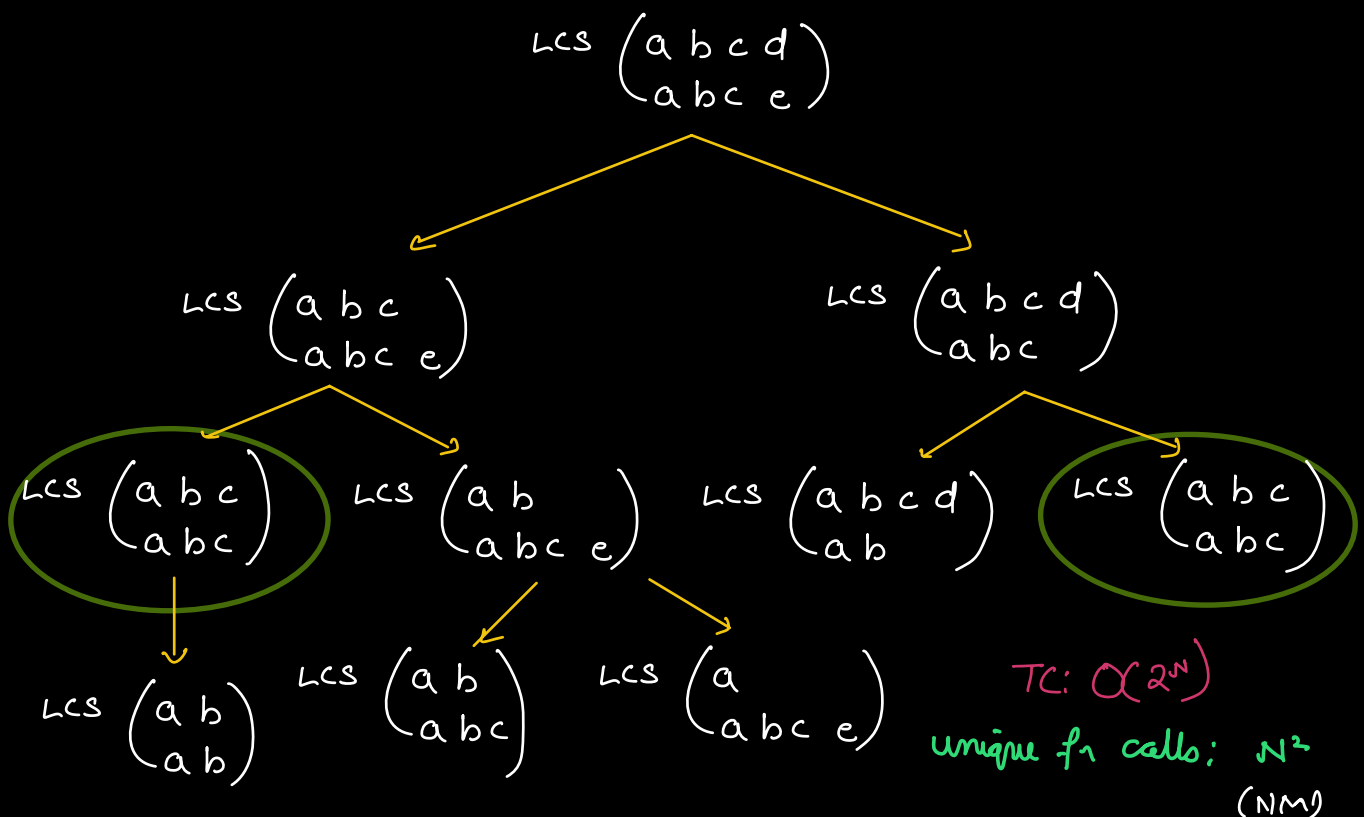
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int LCS (S1, i, S2, j)
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```
if (i < 0 || j < 0)
    return 0;
```

```
if (S1[i] == S2[j]) {
    ret 1 + LCS(S1, i-1, S2, j-1);
}
```

```
ret max (LCS(S1, i-1, S2, j),
        LCS(S1, i, S2, j-1));
```

```
}
```



$DP[N][M] = \{-1\}$

int LCS (s₁, i, s₂, j)

if (i < 0 || j < 0)

ret 0;

if (DP[i][j] == -1) {

if (s₁[i] == s₂[j]) {

DP[i][j] = 1 + LCS(s₁, i-1, s₂, j-1);

}

else {

DP[i][j] = max (LCS(s₁, i-1, s₂, j) ,
LCS(s₁, i, s₂, j-1))

}

}

ret DP[i][j];

}

TC : O(NM)

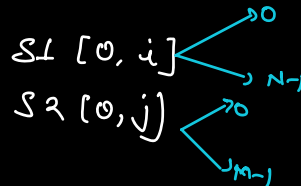
SC : O(NM) + O(N)

Matrix

max(N, M)

rec Stack

DP[i][j] → LCS of s₁[0, i] and s₂[0, j]



Iterative / Bottom-up

$DP[i][j] \rightarrow$ LCS till index i (S_1)
 & index j (S_2)

$DP[N][M]$

$\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ A & B & C & D & E & H \\ A & E & D & F & H & R \end{matrix}$

if ($i < 0$ || $j < 0$)
 ret 0;

		A	B	C	D	E	H
	0	1	2	3	4	5	
→	0	0	0	0	0	0	
A 0	0	1	1	1	1	1	1
E 1	0	1	1	1	1	1	1
D 2	0	1	1	1	2	2	2
F 3	0	1	1	1	2	2	2
H 4	0	1	1	1	2	2	3
R 5	0	1	1	1	2	2	3

← Ans

$$DP[i][j] = \begin{cases} 1 + DP[i-1][j-1] & \text{if } S_1[i] == S_2[j] \\ \max \begin{pmatrix} DP[i-1][j] \\ DP[i][j-1] \end{pmatrix} & \text{else} \end{cases}$$

- $DP[N+1][M+1]$
- $DP[i][0] \rightarrow 0$
- $DP[0][j] \rightarrow 0$
- $DP[i][j]$

$$\text{if } (S_1[i-1] == S_2[j-1])$$

$$DP[i][j] = 1 + DP[i-1][j-1]$$

else

$$DP[i][j] = \max(DP[i-1][j], DP[i][j-1])$$

Q Edit distance

Given 2 strings S_a & S_b

To convert S_a to S_b we have three operations

ops	cost
insert $\longrightarrow$	i
delete $\longrightarrow$	d
update $\longrightarrow$	u

Ret min cost in which S_a can be converted to S_b .

$S_a: ac$

$i: 2$

$d: 2$

$u: 3$

$S_b: abc$

$\Rightarrow 2$

$a \ b \ c$
 $\uparrow$
 $i: 2$

$S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x} \overset{4}{y} \overset{5}{l}$
 $S_b: a \ b \ c \ g \ x \ l$

$\downarrow$

$S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x} \overset{4}{y}$
 $S_b: a \ b \ c \ g \ x$
 $\uparrow$

Del

update

Insert

$S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x} \overset{4}{y}$
 $S_b: a \ b \ c \ g \ x$
 $\uparrow$

$costD + S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x}$
 $S_b: a \ b \ c \ g \ x$

$S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x} \overset{4}{y}$
 $S_b: a \ b \ c \ g \ x$
 $\uparrow$

$costU + S_a: a \ b \ d \ x$
 $S_b: a \ b \ c \ g$

$S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x} \overset{4}{y}$
 $S_b: a \ b \ c \ g \ x$

$costI +$
 $S_a: \overset{0}{a} \overset{1}{b} \overset{2}{d} \overset{3}{x} \overset{4}{y}$
 $S_b: a \ b \ c \ g$

$$ED(s_a, l_a, s_b, l_b) = \begin{cases} ED(s_a, l_a-1, s_b, l_b-1) & \text{if } (s_a[l_a-1] == s_b[l_b-1]) \\ \min \begin{pmatrix} d + ED(s_a, l_a-1, s_b, l_b) \\ u + ED(s_a, l_a-1, s_b, l_b-1) \\ i + ED(s_a, l_a, s_b, l_b-1) \end{pmatrix} & \text{otherwise} \end{cases}$$

Base Conditions

$$\begin{aligned} \text{if } l_a \rightarrow 0 \text{ \& } l_b \rightarrow 0 & : 0 \\ \text{if } l_a \rightarrow 0 & : i \times l_b \\ \text{if } l_b \rightarrow 0 & : d \times l_a \end{aligned}$$

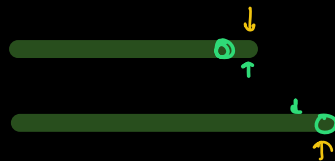
TC

No of unique fn calls / States $\rightarrow O(NM)$

$$\begin{aligned} l_a &\rightarrow [0-N] \\ l_b &\rightarrow [0-M] \end{aligned} \Rightarrow (N+1) \times (M+1)$$

TC: $O(NM)$

SC: $O(NM)$



		s _b →					
		0	1	2	3	4	5
s _a ↓	0	0	i	2i	3i	4i	5i
	1	d					
	2	2d					
	3	3d					
	4	3d					
	5	5d					